LONGEVITYDIET | TARGET DATABASE ARCHITECTURE

DBML SOURCE • 33 LOGICAL ENTITY TYPES • SIX SEPARATE DATABASE OWNERS • ONE VECTOR SHEET

IDENTITY DATABASE

TARGET · LOGICAL SQL SERVER DB | 7 TABLES

Users

PK	Id	uniqueidentifier
	Email	nvarchar(320)
UQ	NormalizedEmail	nvarchar(320)
	PasswordHash	nvarchar(512)
	DisplayName	nvarchar(120)
	Role	nvarchar(24)
	Status	nvarchar(24)
	CreatedAt	datetimeoffset
	UpdatedAt	datetimeoffset

UserProfiles

PK	UserId	uniqueidentifier
	BirthYear	int
	TimeZone	nvarchar(100)
	WakeTime	time
	SleepTime	time
	PreferredMealFrequency	int
	FoodPreference	nvarchar(500)
	ProfileCompleted	bit
	UpdatedAt	datetimeoffset

Reminders

PK	Id	uniqueidentifier
FK	UserId	uniqueidentifier
	Category	nvarchar(40)
	Enabled	bit
	PreferredTime	time
	QuietStart	time
	QuietEnd	time
	UpdatedAtUtc	datetime2

RefreshTokens

PK	Id	uniqueidentifier
FK	UserId	uniqueidentifier
UQ	TokenHash	nvarchar(64)
	ExpiresAt	datetimeoffset
	RevokedAt	datetimeoffset
	ReplacedByTokenId	uniqueidentifier
	CreatedAt	datetimeoffset

UserExcludedFoods

PK	Id	uniqueidentifier
FK	UserId	uniqueidentifier
	FoodId	uniqueidentifier
	FreeText	nvarchar(300)

Allergies

PK	Id	uniqueidentifier
UQ	Code	nvarchar(64)
	Name	nvarchar(200)
	Severity	nvarchar(24)

UserAllergies

FK	UserId	uniqueidentifier
FK	AllergyId	uniqueidentifier
	RecordedAtUtc	datetime2

CATALOG DATABASE

TARGET · LOGICAL SQL SERVER DB | 8 TABLES

Foods

PK	Id	uniqueidentifier
	Name	nvarchar(200)
	NormalizedName	nvarchar(200)
	Category	nvarchar(80)
	Calories	decimal(10,2)
	ProteinG	decimal(9,2)
	CarbohydrateG	decimal(9,2)
	FatG	decimal(9,2)
	IsPlantBased	bit
	IsWholeGrain	bit
	IsLegume	bit
	IsFish	bit
	IsActive	bit

RuleVersions

PK	Id	uniqueidentifier
FK	DietRuleId	uniqueidentifier
	VersionNumber	int
	ParametersJson	nvarchar(max)
	SourceUrl	nvarchar(1000)
	EvidenceLevel	nvarchar(40)
	PublishedAtUtc	datetime2
	IsCurrent	bit

DietRules

PK	Id	uniqueidentifier
UQ	Code	nvarchar(80)
	Name	nvarchar(200)
	Category	nvarchar(80)
	IsMedicalSensitive	bit
	IsActive	bit

Recipes

PK	Id	uniqueidentifier
	Name	nvarchar(200)
	MealType	nvarchar(40)
	Description	nvarchar(1000)
	PrepMinutes	int
	IsPlantForward	bit
	IsActive	bit

RuleSetVersions

PK	Id	uniqueidentifier
	SetName	nvarchar(160)
	Version	int
	PublishedAtUtc	datetime2
	IsCurrent	bit

RecipeIngredients

FK	RecipeId	uniqueidentifier
FK	FoodId	uniqueidentifier
	Amount	decimal(10,2)
	Unit	nvarchar(40)
	IsOptional	bit

RuleSetVersionItems

FK	RuleSetVersionId	uniqueidentifier
FK	RuleVersionId	uniqueidentifier
	Position	int

RecipeAllergens

FK	RecipeId	uniqueidentifier
	AllergenCode	nvarchar(64)
	Evidence	nvarchar(200)

PLANNING DATABASE

TARGET · LOGICAL SQL SERVER DB | 7 TABLES

MealPlans

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	StartDate	date
	EndDate	date
	Revision	int
	Status	nvarchar(40)
	GeneratedAtUtc	datetime2

PlannedMeals

PK	Id	uniqueidentifier
FK	MealPlanDayId	uniqueidentifier
	RecipeId	uniqueidentifier
	MealType	nvarchar(40)
	ScheduledTime	time
	Position	int
	ReasonJson	nvarchar(max)

Challenges

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	Title	nvarchar(200)
	StartDate	date
	EndDate	date
	Status	nvarchar(40)

FmdSafetyAssessments

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	AssessedAtUtc	datetime2
	RiskCode	nvarchar(64)
	IsEligible	bit
	ConsentConfirmed	bit

FmdCycles

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
FK	SafetyAssessmentId	uniqueidentifier
	StartDate	date
	EndDate	date
	Status	nvarchar(40)

ChallengeDays

PK	Id	uniqueidentifier
FK	ChallengeId	uniqueidentifier
	Date	date
	Completed	bit

MealPlanDays

PK	Id	uniqueidentifier
FK	MealPlanId	uniqueidentifier
	Date	date

TRACKING DATABASE

TARGET · LOGICAL SQL SERVER DB | 7 TABLES

AdherenceScores

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	LocalDate	date
	Version	nvarchar(40)
	Score	decimal(5,2)
	CoveragePercent	decimal(5,2)
	IsPartial	bit
	CalculatedAtUtc	datetime2

ActivityLogs

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	ActivityType	nvarchar(40)
	StartedAtUtc	datetime2
	DurationMinutes	int
	Intensity	nvarchar(40)
	IsDeleted	bit

MealLogs

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	ConsumedAtUtc	datetime2
	MealType	nvarchar(40)
	Note	nvarchar(1000)
	IsDeleted	bit

ScoreDimensions

PK	Id	uniqueidentifier
FK	AdherenceScoreId	uniqueidentifier
	Code	nvarchar(40)
	Weight	decimal(5,4)
	RawScore	decimal(5,2)
	WeightedScore	decimal(5,2)
	ReasonCode	nvarchar(100)

MealLogItems

PK	Id	uniqueidentifier
FK	MealLogId	uniqueidentifier
	FoodId	uniqueidentifier
	RecipeId	uniqueidentifier
	Quantity	decimal(9,3)
	Unit	nvarchar(40)

EatingWindowSnapshots

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	LocalDate	date
	WindowStart	time
	WindowEnd	time
	DurationHours	decimal(5,2)

WeeklyReports

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	WeekStart	date
	WeekEnd	date
	SummaryJson	nvarchar(max)
	GeneratedAtUtc	datetime2

RECOMMENDATION DATABASE

TARGET · LOGICAL SQL SERVER DB | 1 TABLES

RecommendationFeedback

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	RequestId	uniqueidentifier
	ItemId	uniqueidentifier
	Rating	int
	ReasonCode	nvarchar(80)
	CreatedAtUtc	datetime2

WORKER DATABASE

TARGET · LOGICAL SQL SERVER DB | 3 TABLES

OutboxMessages

PK	Id	uniqueidentifier
	EventType	nvarchar(120)
	PayloadJson	nvarchar(max)
	CreatedAtUtc	datetime2
	PublishedAtUtc	datetime2
	PublishAttempts	int

AuditLogs

PK	Id	uniqueidentifier
	UserId	uniqueidentifier
	Action	nvarchar(120)
	OccurredAtUtc	datetime2
	CorrelationId	nvarchar(120)
	Result	nvarchar(50)

ProcessedEvents

	EventId	uniqueidentifier
	ConsumerName	nvarchar(120)
	ProcessedAtUtc	datetime2

PK = PRIMARY KEY | FK = FOREIGN KEY (ONLY WITHIN SAME LOGICAL DB) | UQ = UNIQUE | ARROWS = LOCAL RELATIONSHIPS

TARGET OWNERSHIP PROPOSAL · 18 SQL FK links · Cross-service IDs are scalar references, NOT relational joins.